

Question One: Choose the correct answer of the following questions and transfer it in CAPITAL letter to table:

Q.1) All of the orbitals in a given subshell have the same value of the _____ quantum number.
 A) principal B) angular momentum C) magnetic D) A and B E) B and C

Q.2) Consider the following electron configurations to answer the questions that follow:

- (i) $1s^2 2s^2 2p^6 3s^1$
- (ii) $1s^2 2s^2 2p^6 3s^2$
- (iii) $1s^2 2s^2 2p^6 3s^2 3p^1$
- (iv) $1s^2 2s^2 2p^6 3s^2 3p^4$
- (v) $1s^2 2s^2 2p^6 3s^2 3p^5$

The electron configuration that belongs to the atom with the lowest second ionization energy is _____.
 A) (i) B) (ii) C) (iii) D) (iv) E) (v)

Q.3) The formula weight of potassium dichromate ($K_2Cr_2O_7$) is _____ g/mol. (Atomic masses (g/mol): K=39, Cr=52, O=16)

- A) 107 B) 255 C) 242 D) 294 E) 333

Q.4) How many molecules of carbon dioxide are in 308.0 grams of carbon dioxide CO_2 ? ($M.m CO_2 = 44 g/mol$)

- A) 3.499×10^{22} B) 2.107×10^{24} C) 4.214×10^{24} D) 9.274×10^{25} E) 4.081×10^{27}

Q.5) What are the concentrations (M) of Mg^{+2} and $C_2H_3O_2^-$ resulted by dissolving 0.600 mol $Mg(C_2H_3O_2)_2$ in water and diluting to 135 mL?

- A) 0.444 and 0.889 B) 0.044 and 0.089 C) 0.889 and 0.444 D) 0.444 and 0.444 E) 4.440 and 8.890

Q.6) What volume (mL) of a concentrated solution of magnesium chloride (9.00 M) must be diluted to 350. mL to make a 2.75 M solution of magnesium chloride?

- A) 2.75 mL B) 50.0 mL C) 45.0 mL D) 107.0 mL E) 350.0 mL

Q.7) Calculate the mass of $Cu(NO_3)_2$ required to prepare 350.00 mL of a 0.500 M solution? Molar mass of $Cu(NO_3)_2 = 188.0 g/mol$.

- A) 7.83 g B) 14.1 g C) 23.5 g D) 32.9 g E) 47.0 g

Q.8) According to first law of thermodynamics _____

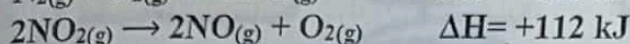
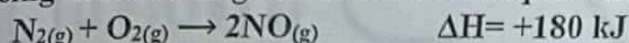
- A) mass and energy are mutually convertible
- B) entropy of a system remains constant.
- C) heat and work are mutually convertible
- D) mass and light are mutually convertible
- E) heat flows from hot substance to cold substance.

Q.9) A 0.85 g gummy sample is burned in a bomb calorimeter. The temperature started at 23.5 °C and leveled off at 27.3 °C. The heat capacity of the calorimeter to be 11.4 kJ/°C. Calculate the heat of combustion per gram of gummy sample.

- A) 34.98 kJ/g
- B) 43.89 kJ/g
- C) 50.96 kJ/g
- D) 59.65 kJ/g
- E) 64.55 kJ/g

Q.10) Calculate ΔH for the reaction: $N_{2(g)} + 2O_{2(g)} \rightarrow 2NO_{2(g)}$ $\Delta H = ?$

Using the following thermochemical equations:

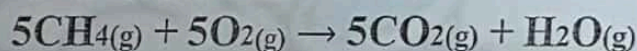


- A) -68.0 kJ
- B) +68.0 kJ
- C) -182.8 kJ
- D) +248.0 kJ
- E) -248.0 kJ

Q.11) A closed system is one in which _____

- A) Mass does not cross boundaries, but energy may do so
- B) Neither mass nor energy crosses the boundaries of the system
- C) Both energy and mass cross the boundaries of the system
- D) Mass crosses the boundary but not the energy
- E) Thermodynamic reactions take place.

Q.12) Determine ΔG° at 25 °C for the reaction:



If ΔG_f° (kJ/mol): $CH_{4(g)} = -50.72$; $H_2O_{(g)} = -228.6$; and $CO_{2(g)} = -394.4$

- A) +101 kJ
- B) -347 kJ
- C) -856.1 kJ
- D) +1479 kJ
- E) -1947 kJ

Q.13) Calculate ΔS° for the reaction: $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$. If S° (J/mol): $Mg_{(s)} = 32.7$; $O_{2(g)} = 205.1$; and $MgO_{(s)} = 26.9$

- A) +146.4 J/mol
- B) -146.4 J/mol
- C) +133.8 J/mol
- D) -216.7 J/mol
- E) +216.7 J/mol

Q.14) The oxidation number of nitrogen in the series, N_2 , O , N_2 , NH_3 , and NO_3^- respectively are:

- A) 0, +2, +5, -3
- B) +1, 0, -3, +5
- C) 0, +1, +5, -3
- D) +5, -3, 0, +1
- E) -3, +5, 0, +1

Q.15) In which one of the following compounds is rhenium (Re) in the highest oxidation state?

- A) $NaReO_4$
- B) $ReClO$
- C) Re_2O_3
- D) $ReCl_5$
- E) ReO_2

Question Two: Write the final answer for the following questions:

- Q.1) At maximum, a d-subshell can hold _____ electrons, and a p-subshell can hold _____ electrons f-subshell can hold _____ electrons.
- Q.2) Atomic radius generally decreases as we move _____.
- Q.3) As successive electrons are removed from an element, the size of atom _____.
- Q.4) Silver nitrate AgNO_3 (M.m = 169.87 g/mol) and aluminum chloride AlCl_3 (M.m = 133.34 g/mol) react with each other by exchanging anions: $3\text{AgNO}_3(\text{aq}) + \text{AlCl}_3(\text{aq}) \rightarrow \text{Al}(\text{NO}_3)_3(\text{aq}) + 3\text{AgCl}(\text{s})$
The mass in grams of AgCl (M.m = 143.32 g/mol) is produced when 5.22 g of AgNO_3 react with 8.73 g of AlCl_3 _____ g.
- Q.5) When aqueous solutions of AgNO_3 and NaCl are mixed, AgCl precipitates. The balanced equation is: _____ $\rightarrow$ _____.
- Q.6) How much heat is required to raise the temperature of an 850 gram block of aluminum from 22.8°C to 84.6°C ? The specific heat of Al is $0.902 \text{ J/g}^\circ\text{C}$. _____ kJ.
- Q.7) Calculate ΔH° for the following reaction:
 $\text{HBr}(\text{g}) \rightarrow \text{H}^+(\text{aq}) + \text{Br}^-(\text{aq}) \quad \Delta H^\circ = ?$
If ΔH_f° , $\text{HBr}(\text{g}) = -36.2 \text{ kJ/mol}$; and ΔH_f° , $\text{Br}^-(\text{aq}) = -120.9 \text{ kJ/mol}$. _____ kJ.
- Q.8) ΔH and ΔE for the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ at constant temperature are related as: _____.
- Q.9) Calculate the Gibbs free energy for a reaction at 273 Kelvin if the change in enthalpy is 19.07 kJ and the change in entropy is 90 J per Kelvin. _____ kJ.
- Q.10) Identify the species being oxidized and reduced in the following reaction:
 $3\text{Hg}^{2+} + \text{Fe}(\text{s}) \rightarrow 3\text{Hg}_2 + 2\text{Fe}^{3+}$ _____.

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